

“Be My Cheese?”: Cultural Nuance Benchmarking for Machine Translation in Multilingual LLMs

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Abstract

We present a large-scale human evaluation benchmark for assessing cultural localisation in machine translation produced by state-of-the-art multilingual large language models (LLMs). Existing MT benchmarks emphasise token-level and grammatical accuracy, but often overlook pragmatic and culturally grounded competencies required for real-world localisation. Building on a pilot study of 87 translations across 20 languages, we evaluate 7 multilingual LLMs across 15 target languages with 5 native-speaker raters per language. Raters scored both full-text translations and segment-level instances of culturally nuanced language (idioms, puns, holidays, and culturally embedded concepts) on an ordinal 0–3 quality scale; segment ratings additionally included an NA option for untranslated segments.

Across full-text evaluations, mean overall quality is modest (1.68/3): GPT-5 (2.10/3), Claude Sonnet 3.7 (1.97/3), and Mistral Medium 3.1 (1.84/3) form the strongest tier with fewer catastrophic failures. Segment-level results show sharp category effects: holidays (2.20/3) and cultural concepts (2.19/3) translate substantially better than idioms (1.65/3) and puns (1.45/3), and idioms are most likely to be left untranslated. These findings demonstrate a persistent gap between grammatical adequacy and cultural resonance. To our knowledge, this is the first multilingual, human-annotated benchmark focused explicitly on cultural nuance in translation and localisation, highlighting the need for culturally informed training data, improved cross-lingual pragmatics, and evaluation paradigms that better reflect real-world communicative competence.

1 Introduction

Large language models (LLMs) have rapidly expanded access to machine translation, enabling rapid translation across hundreds of languages without requiring linguistic expertise. Cultural nuances, such as figurative expressions and idioms, are foundational to effective human communication and shape how meaning is received and interpreted by local audiences. A translation that is grammatically correct may nevertheless sound unnatural, inappropriate, or misleading if it fails to account for cultural context. However, machine translation (MT) research and benchmarks continue to prioritise lexical and grammatical accuracy at the token- and sentence-level. These metrics capture formal correctness, but fail to evaluate the pragmatic, cultural, and stylistic competencies required for real-world localisation tasks such as marketing communication, customer engagement, and culturally specific brand messaging.

This study introduces a benchmark designed explicitly for evaluating how well multilingual LLMs preserve cultural resonance in machine translation tasks. Building on a pilot evaluation of 87 translations across 20 languages (Anonymous, 2025), we scale to a substantially larger dataset comprising 7 state-of-the-art multilingual LLMs, 15 target languages, and five native-speaker raters per language. Each rater evaluated both (1) a complete translated marketing email and (2) pre-defined segment-level instances of culturally nuanced language, including idioms, puns, holiday references, and culturally embedded concepts. This design allows us to contrast holistic translation quality with categorical failure modes on a phrasal level.

Our study addresses three core research questions:

- How well do contemporary multilingual LLMs translate culturally nuanced language across typologically diverse languages?
- To what extent do model family, linguistic characteristics, and orthographic systems impact cultural resonance in MT?
- Which categories of culturally marked content, such as idioms and puns, pose the greatest challenges to current LLMs?

Our findings reveal a substantial gap between grammatical accuracy and cultural localisation. While many translations achieve surface-level adequacy, even the strongest multilingual LLMs fail to consistently preserve culturally grounded meaning, particularly for figurative and non-literal language. These results underscore the limitations of existing machine translation in SOTA models and motivate a reevaluation of MT benchmarks and training practices that prioritises cultural-pragmatic competence as a core dimension of multilingual LLM performance.

1.1 Contributions

This work presents three primary contributions:

1. A new benchmark for culturally sensitive machine translation.

We introduce the first multilingual, human-annotated benchmark designed explicitly to evaluate cultural nuance and resonance in machine translation, spanning 7 state-of-the-art multilingual LLMs, 15 languages, and five native-speaker raters per language, with both holistic and segment-level evaluation.

2. A large-scale empirical analysis of cultural failure modes in MT.

Through segment-level evaluation of idioms, puns, holidays, and culturally embedded concepts, we show that cultural localisation quality diverges sharply from grammatical accuracy, with figurative language remaining a persistent failure mode across models and languages.

3. Evidence of systematic model- and language-level variation in cultural MT performance.

We identify consistent performance differences across models, languages, and orthographic systems, including higher stability among GPT-5, Claude Sonnet 3.7, and Mistral Medium 3.1, and elevated failure rates for culturally marked segments in other systems, motivating targeted data and evaluation strategies for improving cultural competence in multilingual LLMs.

2 Related Work

Recent advances in large language models (LLMs) have driven substantial improvements in multilingual machine translation. Mujadia et al. (2023) provide a comprehensive assessment of LLM translation performance between English and 22 Indian languages, revealing persistent disparities across high- and low-resource settings and demonstrating the benefits of in-context learning for underrepresented dialects. Similarly, Hu et al. (2024) introduce GenTranslate, showing that generative LLM-based approaches improve multilingual speech and text translation on standard benchmarks, particularly for low-resource languages. Together, these studies illustrate rapid progress in multilingual MT while highlighting uneven gains across languages.

Despite these advances, most prior evaluations focus on lexical and grammatical accuracy, relying on automatic metrics or sentence-level adequacy judgments. Such evaluations are poorly suited to capturing pragmatic and cultural dimensions of translation quality, including idiomatic meaning, figurative language, and audience-appropriate tone. As a result, translations that are formally correct may nevertheless be culturally inappropriate or misleading in real-world localisation contexts. This limitation is well documented in the MT evaluation literature. BLEU has long been shown to correlate weakly with meaning adequacy and human judgments beyond surface correspondence (Callison-Burch et al., 2006; Mathur et al., 2020), and more recent neural metrics such as COMET and BLEURT similarly struggle with discourse-level, pragmatic, and culturally grounded errors (Freitag et al., 2021; Kocmi et al., 2022). Recent work has therefore begun to frame cultural transfer and adaptation as a core challenge for language technologies, arguing that culture-aware evaluation is necessary to capture meaning beyond surface correspondence (Singh et al., 2024).

Beyond translation accuracy, a growing body of work has examined cultural alignment in LLM outputs more broadly. AlKhamissi et al. (2024) investigate cultural alignment across languages and regions, showing that LLMs better reflect culturally grounded knowledge when prompted in a region’s dominant language, while also identifying persistent representation gaps. Li et al. (2024) propose CultureLLM, incorporating culturally diverse multilingual data to improve cultural appropriateness in generation tasks. While these approaches demonstrate measurable gains, they largely focus on open-ended generation rather than translation and do not systematically evaluate how well models preserve culturally meaningful content when transferring meaning across languages.

The present work builds most directly on a pilot study by Anonymous (2025), which evaluated 87 translations across 20 languages and found that figurative language posed a consistent challenge even for high-performing models. While the pilot demonstrated the limitations of existing MT benchmarks for real-world localisation, it was constrained in scale and statistical power. The current study substantially extends this work by evaluating seven state-of-the-art multilingual LLMs across fifteen languages with multiple native-speaker raters per language, introducing segment-level evaluation of culturally nuanced language, and applying statistical modeling to disentangle the effects of model, language, and content type.

By situating cultural nuance as a core dimension of translation quality rather than a peripheral concern, this work complements existing MT and cultural alignment research and addresses a critical gap in current evaluation paradigms for multilingual LLMs.

3 Methodology

We evaluate multilingual LLMs on their ability to translate and culturally localise English marketing emails into 15 target languages. Unlike traditional MT benchmarks that emphasise lexical and grammatical accuracy, this task requires models to handle culturally marked language, including idioms, puns, holiday references, figurative expressions, and culturally embedded concepts.

Each model received the same English source text and a fixed prompt instructing it to “*Translate the following email for use in [language] in [country/region].*” All translations were generated in fresh chat sessions to minimise contamination across runs.

3.1 Languages and Participants

We recruited five native speakers per language (N = 75 total) across 15 locales: Afrikaans (ZA), Arabic (EG), Brazilian Portuguese (BR), Cantonese (HK), Czech (CZ), Dutch (NL), Hebrew (IL), Hindi (IN), Japanese (JP), Korean (KR), Mandarin (TW), Russian (KZ), Spanish (MX), Swahili (KE), and Urdu (PK).

Participants reside in the region they evaluated and are fluent speakers of both English and their native language. Each rater evaluated translations only for their native language. Participant demographic information is provided in Appendix B1.

3.2 Models Evaluated

We evaluate seven publicly available multilingual LLMs including a range of leading developers as well as open- and closed-weight systems.

Developer	Model	Weight Type
Anthropic	Claude Sonnet 3.7	Closed-weight
Mistral	Medium 3.1	Closed-weight
DeepSeek	V3.1	Closed-weight
OpenAI	GPT-5	Closed-weight
OpenAI	gpt-oss 120B	Open-weight
Meta	Llama 4	Open-weight
Cohere	Aya Expanse 8B	Open-weight

Table 1 List of models evaluated.

When models produced meta-comments or explanations, English text was removed prior to evaluation. Non-English explanatory text was retained only when it was inseparable from the translated output.

3.3 Input Materials

Source texts consisted of five e-commerce marketing emails adapted from authentic commercial campaigns. These emails were selected because they contain:

- puns and humorous wordplay
- holiday-specific phrases
- idiomatic expressions
- culturally specific references
- strong brand voice and audience targeting

From each email, we selected five segments of culturally nuanced language. Across the dataset, this resulted in four puns, four idioms, four holiday references, and thirteen cultural concepts per language. Cultural concepts were defined as single words or short phrases that are either specific to North American English or unlikely to have direct equivalents across cultures (e.g., *koozies*, *sweetheart*, *zero-waste*). Full source texts and segment selections are provided in Appendix A.

3.4 Evaluation Procedure

Each rater assessed one translation per model, evaluating both the full translated text and segments. Full participant guidelines are presented in Appendix B2.

(a) Full text evaluation

Participants scored the translation on a 4-point scale for the following criteria:

1. Content fidelity
2. Style fidelity
3. Audience appropriateness
4. Overall translation quality

These items measure whether the translation is correct, natural, locally resonant, and aligned with the original intent. Participants were also given free response text boxes to provide additional qualitative feedback. A summary of the qualitative feedback by language is available in Appendix D.

(b) Segment-level evaluation

Raters also evaluated predefined culturally nuanced segments from the emails, each labeled as one of:

- idioms
- puns
- holidays
- cultural concepts

Segments were rated on the same 0–3 scale, with an additional NA option indicating the segment was not translated, instead opting to retain the original English. This enables fine-grained

analysis of where models succeed or fail in cultural MT beyond full-text impressions. This methodology produced 13,125 segment-specific annotations.

3.5 Annotation Protocol

Participants received detailed written instructions based on an evaluation framework (available in Appendix B2), including:

- definitions of cultural nuance
- examples of literal vs. localised translation strategies
- guidance on how to rate ambiguous cases
- clarifications for rating idioms and humour

Ratings were collected using our proprietary data annotation software (redacted for anonymity). Each submission was checked for completeness and annotation consistency.

3.6 Statistical Analysis

We analysed segment-level translation ratings using a cumulative link mixed model (CLMM) with a logit link, appropriate for ordinal outcomes. Models were fitted in R using the ordinal package (Christensen, 2022). Fixed effects included model, language, and segment category, as well as their interaction. Random intercepts were included for annotator and segment to account for repeated ratings and item-level variability.

Orthography was initially included as a fixed effect but was removed from the final specification due to rank deficiency and near-complete collinearity with language–category combinations. Its inclusion resulted in unstable parameter estimates without improving model fit. The final model converged successfully ($\log\text{Lik} = -14,411.63$; $\text{AIC} = 28,965.26$; $n = 13,125$). Random-effects estimates indicate greater variance at the segment level ($\text{SD} = 1.76$) than at the annotator level ($\text{SD} = 0.70$), suggesting that segment-specific difficulty contributes more to rating variability than individual rater severity.

Inter-rater reliability (IRR) was assessed separately for full-text (overall) ratings and segment-level ratings using Krippendorff’s α (ordinal) and Gwet’s AC2 with quadratic weights. IRR was computed overall and stratified by model, language, and segment category. Full-text IRR

assesses consistency in holistic translation judgments, while segment-level IRR evaluates agreement on fine-grained, culturally marked language. Ratings corresponding to “segment not translated” were excluded from IRR analyses.

Full model specifications, IRR tables, and post-hoc comparisons are reported in Appendix C.

4 Results

We report results from both holistic full-text evaluation and segment-level evaluation of culturally nuanced language. All scores are reported on a 0–3 ordinal scale, where higher values indicate better translation quality.

4.1 Full-Text Translation Quality by Model

Full-text translation quality remains modest overall (mean = 1.68/3). Descriptive averages place GPT-5 (2.10/3), Claude Sonnet 3.7 (1.97/3), and Mistral Medium 3.1 (1.84/3) at the top of the distribution, with Aya Expanse 8B substantially lower (1.09/3). Table 2 summarises average full-text scores by model.

Model	overall quality	audience	style	content
Claude Sonnet 3.7	1.97	2.25	2.08	2.10
Cohere Aya Expanse 8B	1.09	1.55	1.41	1.21
DeepSeek V3.1	1.72	2.05	1.98	1.77
GPT-5	2.10	2.38	2.23	2.23
gpt-oss 120B	1.60	1.94	1.83	1.72
Llama 4	1.47	1.81	1.72	1.59
Mistral Medium 3.1	1.84	2.19	2.04	1.92
Total	1.68	2.023	1.90	1.79

Table 2 Average rating on a 0–3 (4-point) ordinal scale by model across languages of overall translation quality, appropriateness to intended audience, faithfulness to style of the original, and faithfulness to content of the original.

CLMM results confirm a significant main effect of model on translation quality (Table C1). Relative to GPT-5, Aya Expanse 8B exhibits markedly worse performance ($\beta = 1.90$, $p < .001$). Llama 4, gpt-oss 120B, and DeepSeek V3.1 also perform significantly worse than GPT-5, while differences between GPT-5, Claude Sonnet 3.7, and Mistral Medium 3.1 are not statistically significant.

Estimated marginal means and Tukey-adjusted comparisons (Tables C2–C3) place GPT-5, Claude Sonnet 3.7, and Mistral Medium 3.1 in a statistically indistinguishable top tier, followed by a middle tier of DeepSeek V3.1 and gpt-oss 120B. Aya Expanse 8B is a clear outlier, performing significantly worse than all other models.

Inter-rater reliability for full-text ratings indicates moderate agreement across models and languages, supporting the stability of the observed model-level effects (Table C9). We report IRR to contextualize the subjectivity of cultural judgments while model and category effects are interpreted primarily through the CLMM estimates and post-hoc comparisons.

4.2 Segment-Level Performance by Category

Segment category exhibits the strongest and most consistent effect on translation quality. CLMM estimates show large and highly significant differences across categories (Tables C6–C7). Holidays and culturally embedded concepts receive substantially higher ratings than idioms and puns ($p < .001$ for all figurative vs. non-figurative contrasts), while the difference between idioms and puns is not statistically significant.

Descriptively, holidays (2.20/3) and cultural concepts (2.19/3) achieve the highest average quality among translated segments, whereas idioms (1.65/3) and puns (1.45/3) perform substantially worse. These results indicate that figurative and non-literal language remains a persistent challenge even when models attempt a translation.

Translation coverage also varies markedly by category. Idioms are most frequently left untranslated (rated NA), followed by puns, while holidays and cultural concepts are more consistently rendered. These omission patterns are reported descriptively and are not included in the CLMM, which models translation quality conditional on a translation being produced.

Segment-level IRR exhibit greater variability in inter-rater agreement, with lower agreement for idioms and puns than for holidays and cultural concepts, reflecting greater annotator uncertainty when evaluating figurative language (Table C8).

4.3 Model Effects on Segment Translation

Controlling for language and segment category, model choice significantly affects segment-level translation quality (Table C1). GPT-5 and Claude Sonnet 3.7 do not differ significantly and outperform gpt-oss 120B, Llama 4, and Aya Expanse 8B. Mistral Medium 3.1 performs significantly better than Aya Expanse 8B and Llama 4, but does not differ significantly from DeepSeek V3.1 or GPT-5.

Aya Expanse 8B is a clear outlier, exhibiting both significantly lower quality scores and substantially higher omission rates for idioms and puns. Other models omit fewer segments overall but frequently produce low-quality translations (ratings 0–1) for figurative language.

IRR stratified by model (Table C8) indicates moderate agreement for GPT-5 and Claude Sonnet 3.7, with greater variability for lower-performing models, suggesting that inconsistent output quality contributes to annotator disagreement.

4.4 Language-Level Effects

Language effects are present but more constrained than category or model effects. CLMM estimates indicate that Mandarin (Taiwan) receives significantly higher segment-level ratings than several other languages, including Spanish, Swahili, and Urdu (Tables C4–C5). Brazilian Portuguese trends higher but does not consistently differ from other languages after correction for multiple comparisons.

Importantly, language effects interact with segment category. Languages that perform well overall tend to maintain higher scores across categories, while lower-performing languages exhibit disproportionate degradation on idioms and puns. This pattern persists even when restricting analysis to translated segments, indicating that low scores are not driven solely by omission.

IRR varies substantially by language (Table C8), with lower agreement for Mandarin, Hindi, and Urdu, suggesting that cultural interpretation differences may amplify annotator variability in these contexts.

5 Discussion

The CLMM analysis confirms that cultural localization failures in multilingual LLMs are systematic rather than anecdotal. Segment category emerges as the strongest predictor of translation quality, exceeding the influence of both model family and language. Figurative language, especially idioms and puns, remains a robust failure mode even after controlling for rater effects and segment-level difficulty.

Crucially, the statistical results support a distinction between translation coverage and translation quality. Idioms are significantly more likely to be omitted entirely, and when translated, they receive substantially lower ratings than holidays or culturally embedded concepts. Aya Expanse 8B exhibits both the highest omission rates and the lowest quality scores for idiomatic translation, indicating that failure is not merely a consequence of conservative behavior. Even when models attempt figurative translation, pragmatic and culturally appropriate rendering frequently fails.

Model-level effects reveal a stable top tier (GPT-5, Claude Sonnet 3.7, and Mistral Medium 3.1) but no system consistently achieves high performance across all categories. The absence of statistically significant differences among these models suggests that scaling and architectural refinement alone do not resolve cultural–pragmatic limitations. In contrast, Aya Expanse 8B’s consistently poor performance across analyses points to systemic fragility rather than isolated weaknesses.

Language-level effects are present but secondary, and orthography does not independently predict translation quality once language and segment category are accounted for. This finding contrasts with observations from the pilot study and challenges assumptions that script or typological complexity are primary drivers of cultural MT difficulty. Instead, the results point toward the availability and quality of culturally situated training data as a more plausible explanation for observed disparities.

6 Future work

Future work will extend this benchmark in several directions. First, we plan to release the dataset and evaluation framework as a public benchmark,

enabling reproducible research on cultural localisation in machine translation and multilingual LLM evaluation. The release will include full-text translations, segment-level annotations, and detailed evaluation guidelines to support consistent comparison across future models. Rather than replacing automatic metrics, this benchmark will complement them by targeting pragmatic and cultural dimensions that current form-based evaluations systematically overlook.

Second, we plan to expand the benchmark beyond text-only translation by developing an audio-based version of the task. Many culturally marked expressions – such as humour, idioms, and tone – are realised differently in spoken language, and evaluating speech-based localisation will allow analysis of dialect, prosody, emphasis, and pragmatic delivery not captured in text. We also intend to extend coverage to additional domains and languages to assess the generality of the cultural failure modes identified here.

7 Conclusion

We presented a large-scale, human-annotated benchmark designed to evaluate cultural localisation in machine translation by multilingual LLMs. Across seven state-of-the-art models and fifteen languages, results reveal a persistent gap between grammatical adequacy and cultural resonance. While many translations appear superficially plausible, segment-level evaluation exposes systematic failures, particularly idioms and puns, that remain largely invisible to standard MT metrics.

By explicitly distinguishing between translation coverage and translation quality, this work provides a more nuanced account of cultural MT performance and highlights limitations shared even by the strongest current models. These findings underscore the need for culturally informed training data and evaluation paradigms that move beyond form-based correctness toward real-world communicative competence.

8 Limitations

This study has several limitations. The benchmark focuses on English-to-many translation within a marketing email domain, which may limit generalisability to other genres such as news, legal text, or conversational dialogue. Segment selec-

tion intentionally emphasises culturally marked language and is therefore not representative of typical sentence distributions in MT corpora. Furthermore, analysis of segment-level MT was performed in the context of larger MT corpora and may not generalise to MT performance when translating the same segments as isolated text. Future work could rectify this limitation by evaluating and contrasting segment-level MT in context of larger text with segment MT as isolated input.

Although five native raters per language reduce individual bias, judgments of cultural appropriateness remain inherently subjective and may vary across demographics, regions, and personal experience within a language community. Additionally, this study did not control for differences in participant age, education, gender, and socioeconomic background – all factors known to influence human bias (Jenks, 2025; Zahraei and Emami, 2025). In addition, models were evaluated through publicly accessible interfaces, which may introduce uncontrolled variation due to system prompts, safety filters, or model updates. Furthermore, model outputs were collected over the span of two days, introducing additional potential for uncontrolled variation when compared to simultaneous output generation and collection. Finally, this work focuses exclusively on text-based translation and does not address multimodal or spoken localisation, which we leave to future research.

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Appendix

Appendix A. Complete MT Input Texts

A1

Company: Sheffield's – a gourmet market in NYC

Subject: Will you brie mine? 🧀❤️🧀

Valentine's Day is almost here, and we've got the sweetest gift ideas for pickup or delivery throughout NYC.

Cheese Tasting Gift Boxes

This cheese lover's dream is thoughtfully assembled by our expert cheesemongers. It all comes beautifully packaged in a keepsake tin, tied with a satin ribbon. Personalize it with a custom note on Sheffield's stationery.

Sweets for your Sweetheart

Artfully displayed with the perfect accompaniments of fresh & dried fruit, nuts, honey, fig jam, espresso brownies, dark chocolate-covered strawberries, candies, edible flowers and sliced baguette.

[order here]

We still have a limited number of handmade, chocolate-covered strawberries and floral arrangements available for pre-order! Give us a call today or stop by the shop before they're gone.

Wishing you a sweet Valentine's Day!

Sheffield's – Park Slope

Brooklyn, NY

A2

Company: Terra – an eco-friendly deodorant brand

Subject: This scent will transform your life ✨

Hey [NAME]

Your New Year's resolution stinks. Give your life

a scent-sational upgrade – pair our newest reusable case design with a fragrance that's sure to make memories. Durable, stylish, compact, and zero waste.

Swipe right this New Year's Eve

Use code: NYE2026

[shop deodorant]

Whether you're keeping yourself fresh for your partner, or looking to impress someone else, our new scents will leave a lasting impression.

MIX & MATCH OUR BEST-SELLING COMBOS

Lavender case x Tropical Paradise scent

Turquoise case x Orange Creamsicle scent

WHY TERRA?

Aluminum & paraben free. Zero-waste refills. 24-hour odor protection. All that in a case you'll be excited to reuse.

Terra Cosmetics

London N1C 4AB, United Kingdom

A3

Company: Muggable – an American novelty mug company

Subject: This Collection Has Us Feline Good 🐱

CAT'S MEOW

Our newest collection is the cat's pajamas, wait no – it's the cat's Mugs, Tumblers, Koozies, and Coasters!

[Shop Meow]

Rep your favorite feline at the office, on the go, and on your next Zoom call. Wait. Who are we kidding? They're already in all your Zoom calls.

© 2012 Muggable Inc. All Rights Reserved.
Los Angeles, CA, 90013, USA

A4

Company: sonia summerhouse– an american luxury swimwear brand

Subject: late Summer, full throttle

Labor Day is here! PACK YOUR BEACH BAG!

you sprint barefoot across warm sand.

the sun hits high.

salt hangs in the air.

seagulls cut the wide blue sky.

laughter bursts, waves crash in time,

summer comes alive.

your new swimwear, green like sea glass.

fabric flowing, grab your crew,

chase the surf, leap, sprint, splash -

shore enough, this is your moment!

Sonia Summerhouse 20 w. 20th street unit 1004
new york, ny 10011

A5

company: Cinnamon – a neighborhood bakery & cafe

Subject: Happy Birthday! There's a sweet treat waiting for you!

Sugar, spice, and everything nice! Happy Birthday from all of us at Cinnamon!

Let us be the icing on the cake of your special day with a sweet treat. Stop by any Cinnamon location to redeem your credit on your next order OR save it for later by visiting the Rewards section in your app.

We can't wait to celebrate with you! Redeemable with the Cinnamon app only.

Excited about your birthday present?
Say Thanks on Facebook

A6 Segments

Segment	Segment category
birthday present	cultural concepts
cheesemongers	cultural concepts
full throttle	cultural concepts
grab your crew	cultural concepts
Happy Birthday	cultural concepts
keepsake tin	cultural concepts
Koozies	cultural concepts
summer comes alive	idioms
sweet treat	cultural concepts
Sweetheart	cultural concepts
Swipe right	cultural concepts
Tumblers	cultural concepts
Zero-waste	cultural concepts
Zoom call	cultural concepts
New Year's Eve	holidays
NYE2026	holidays
Labor Day	holidays
Valentine's Day	holidays
cat's pajamas	idioms
icing on the cake	idioms
Sugar, spice, and everything nice	idioms
Feline Good	puns
scent-sational	puns
shore enough	puns
Will you brie mine?	puns

Table A6 Segmentation and categorisation of phrases and words selected for individual evaluation.

Appendix B. Participants

B1 Participant Demographics

Language	Participant age	Participant Gender	Participant education level
Afrikaans	31-45	female	Secondary education completed (high school diploma or equivalent)
Afrikaans	31-45	female	Postgraduate diploma or certificate (non-degree)
Afrikaans	31-45	female	Postgraduate diploma or certificate (non-degree)
Afrikaans	31-45	female	Some college or university (no degree)
Afrikaans	45+	female	Some college or university (no degree)
Arabic	45+	female	Bachelor's degree (e.g., BA, BS)
Arabic	18-30	male	Bachelor's degree (e.g., BA, BS)
Arabic	31-45	male	Master's degree (e.g., MA, MS, MBA, MFA)
Arabic	31-45	female	Bachelor's degree (e.g., BA, BS)
Arabic	18-30	male	Bachelor's degree (e.g., BA, BS)
Brazilian Portuguese	18-30	male	Secondary education completed (high school diploma or equivalent)
Brazilian Portuguese	31-45	female	Some college or university (no degree)
Brazilian Portuguese	45+	male	Master's degree (e.g., MA, MS, MBA, MFA)
Brazilian Portuguese	31-45	male	Master's degree (e.g., MA, MS, MBA, MFA)
Brazilian Portuguese	45+	male	Postgraduate diploma or certificate (non-degree)
Cantonese	31-45	female	Bachelor's degree (e.g., BA, BS)
Cantonese	18-30	female	Bachelor's degree (e.g., BA, BS)
Cantonese		UNKNOWN	Bachelor's degree (e.g., BA, BS)
Cantonese	18-30	female	Bachelor's degree (e.g., BA, BS)
Cantonese	45+	female	Bachelor's degree (e.g., BA, BS)
Czech	31-45	female	Bachelor's degree (e.g., BA, BS)
Czech	18-30	female	Master's degree (e.g., MA, MS, MBA, MFA)
Czech	18-30	male	Some secondary education (high school)
Czech	18-30	male	Some college or university (no degree)
Czech	31-45	male	Master's degree (e.g., MA, MS, MBA, MFA)
Dutch	31-45	male	Bachelor's degree (e.g., BA, BS)
Dutch	31-45	male	Bachelor's degree (e.g., BA, BS)
Dutch	45+	female	Bachelor's degree (e.g., BA, BS)
Dutch	31-45	male	Bachelor's degree (e.g., BA, BS)
Dutch	45+	male	Bachelor's degree (e.g., BA, BS)
Hebrew	45+	male	Bachelor's degree (e.g., BA, BS)
Hebrew	31-45	male	Bachelor's degree (e.g., BA, BS)
Hebrew	31-45	female	Bachelor's degree (e.g., BA, BS)
Hebrew	31-45	male	Vocational/technical training or certification (e.g., trade school)
Hebrew	45+	male	Bachelor's degree (e.g., BA, BS)
Hindi	31-45	male	Bachelor's degree (e.g., BA, BS)

Hindi	31-45	male	Doctoral or professional degree (e.g., PhD, MD, JD, PsyD, EdD)
Hindi	45+	male	Master's degree (e.g., MA, MS, MBA, MFA)
Hindi	18-30	male	Postgraduate diploma or certificate (non-degree)
Hindi	18-30	male	Bachelor's degree (e.g., BA, BS)
Japanese	45+	female	Bachelor's degree (e.g., BA, BS)
Japanese	31-45	male	Master's degree (e.g., MA, MS, MBA, MFA)
Japanese	31-45	male	Bachelor's degree (e.g., BA, BS)
Japanese	45+	male	Bachelor's degree (e.g., BA, BS)
Japanese	18-30	male	Bachelor's degree (e.g., BA, BS)
Korean	45+	female	Bachelor's degree (e.g., BA, BS)
Korean	31-45	female	Master's degree (e.g., MA, MS, MBA, MFA)
Korean			Bachelor's degree (e.g., BA, BS)
Korean	31-45	female	Master's degree (e.g., MA, MS, MBA, MFA)
Korean	45+	female	Bachelor's degree (e.g., BA, BS)
Mandarin	31-45	female	Bachelor's degree (e.g., BA, BS)
Mandarin	31-45	female	Bachelor's degree (e.g., BA, BS)
Mandarin	45+	male	Master's degree (e.g., MA, MS, MBA, MFA)
Mandarin	18-30	male	Bachelor's degree (e.g., BA, BS)
Mandarin	31-45	male	Master's degree (e.g., MA, MS, MBA, MFA)
Russian	31-45	male	Master's degree (e.g., MA, MS, MBA, MFA)
Russian	31-45	female	Bachelor's degree (e.g., BA, BS)
Russian	45+	male	Secondary education completed (high school diploma or equivalent)
Russian	45+	female	Bachelor's degree (e.g., BA, BS)
Russian	31-45	male	Master's degree (e.g., MA, MS, MBA, MFA)
Spanish	31-45	female	Bachelor's degree (e.g., BA, BS)
Spanish	31-45	female	Bachelor's degree (e.g., BA, BS)
Spanish	18-30	female	Master's degree (e.g., MA, MS, MBA, MFA)
Spanish	31-45	FEMALE	Some college or university (no degree)
Spanish	31-45	male	Bachelor's degree (e.g., BA, BS)
Swahili	18-30	female	Bachelor's degree (e.g., BA, BS)
Swahili	31-45	female	Postgraduate diploma or certificate (non-degree)
Swahili	18-30	female	Bachelor's degree (e.g., BA, BS)
Swahili	18-30	male	Bachelor's degree (e.g., BA, BS)
Swahili	18-30	male	Bachelor's degree (e.g., BA, BS)
Urdu	31-45	male	Master's degree (e.g., MA, MS, MBA, MFA)
Urdu	31-45	female	Master's degree (e.g., MA, MS, MBA, MFA)
Urdu	31-45	female	Master's degree (e.g., MA, MS, MBA, MFA)
Urdu	18-30	male	Master's degree (e.g., MA, MS, MBA, MFA)
Urdu	31-45	male	Bachelor's degree (e.g., BA, BS)

B2 Participant Guidelines

Overview

This project is meant to evaluate the quality of translation and localization of various LLM models when asked to translate marketing emails from English to a given language and locale. Imagine that a person working at an advertising agency is asked to translate a marketing email they are working on from English to a language and country that they don't know anything about. They do what people do these days and go to the internet and ask their favorite LLM model to "Translate this email into {{language}} for use in {{country}}"

You represent their end user, as a person in the targeted country who speaks the language, we are asking for. We'd like you to evaluate the email from the perspective of a person getting that advertisement in your email. How well is it translated? How well does it target local traditions and norms? How true to the original content and tone is the translation?"

We'll ask several questions, all using the same basic evaluation scale. Keep these ratings and descriptions in mind while you are evaluating.

- serious failures exist - use this in cases where you would be very disappointed, confused, offended, or in some other way have negative feelings towards the company or product because of the content of the translation
- imperfect but not terrible - there are errors or issues that are very noticeable, but that are not so bad as to give a negative impression of the company, the main ideas come through and it is clear what is being advertised.
- mostly good with small issues - the wording or translations are noticeably non-native, or are awkward or a little odd, but it is overall something that makes sense and could be used without embarrassment on the part of the company.
- very good or nearly perfect - this is for something that seems very close to natural, native, and culturally appropriate.

Steps

1. Read both emails.
2. Answer the holistic questions on the left hand side

3. Answer the segment specific questions on the right hand side
4. Give an overall evaluation, considering your ratings both for the holistic and the segment specific ratings
5. Leave free-text comments at any point if you notice something interesting or want to add context to your rating.

Notes

- The translated email may include notes from the model on the translation. Please disregard these and evaluate the translated email from the perspective of a potential customer receiving it in your inbox.

Appendix C. Statistical Modelling

C.1 Cumulative Link Mixed Model Specification

A cumulative link mixed model (CLMM) with a logit link function was fitted to the ordinal translation quality ratings using the ordinal package in R (Christensen, 2022). The model was specified to account for the ordered nature of the response variable while incorporating both fixed and random effects to capture systematic variation across experimental factors and repeated measurements.

Orthography was initially included as a fixed effect; however, preliminary diagnostics indicated that its inclusion resulted in a rank-deficient design matrix, with multiple coefficients automatically dropped during estimation. Inspection of the data revealed sparse cell counts and near-complete collinearity between orthography, language, and segment category. Under these conditions, orthography effects could not be uniquely identified and impeded stable estimation without improving model fit. Orthography was therefore excluded from the final model to preserve identifiability and convergence.

The final fixed-effects structure included model, language, segment category, and their interaction. Random intercepts were specified for annotator and segment to account for repeated ratings by the same individuals and shared difficulty across evaluated segments. Model parameters were estimated via maximum likelihood using the regularized Newton–Raphson algorithm implemented in ordinal.

Predictor	Estimate	SE	CI	z_ratio	p_value	Significance
Very good / nearly perfect Mostly good	-0.01	0.36	[-0.71, 0.69]	-0.03	0.975	
Mostly good Imperfect	1.28	0.36	[0.58, 1.99]	3.57	< .001	***
Imperfect Serious failures	2.50	0.36	[1.80, 3.21]	6.95	< .001	***
Serious failures Segment not translated	6.27	0.37	[5.54, 6.99]	16.92	< .001	***
modelCohere Aya Expanse 8B	1.90	0.15	[1.60, 2.20]	12.42	< .001	***
modelDeepSeek V3.1	0.51	0.15	[0.20, 0.81]	3.28	0.001	**
modelGPT-5	0.02	0.16	[-0.28, 0.33]	0.15	0.878	
modelgpt-oss 120b	0.81	0.15	[0.50, 1.11]	5.25	< .001	***
modelLlama 4	1.03	0.15	[0.72, 1.33]	6.68	< .001	***
modelMistral Medium 3.1	0.38	0.15	[0.08, 0.69]	2.48	0.013	*
languageArabic	0.22	0.48	[-0.72, 1.15]	0.45	0.652	
languageBrazilian Portuguese	-0.92	0.48	[-1.87, 0.03]	-1.89	0.058	
languageCantonese	-0.22	0.48	[-1.15, 0.72]	-0.45	0.650	
languageCzech	0.12	0.48	[-0.81, 1.05]	0.25	0.800	
languageDutch	0.29	0.48	[-0.64, 1.23]	0.62	0.538	
languageHebrew	0.16	0.48	[-0.77, 1.10]	0.34	0.735	
languageHindi	0.60	0.47	[-0.32, 1.53]	1.28	0.202	
languageJapanese	-0.29	0.48	[-1.23, 0.64]	-0.61	0.540	
languageKorean	0.14	0.48	[-0.81, 1.09]	0.29	0.771	
languageMandarin	-1.53	0.49	[-2.50, -0.56]	-3.09	0.002	**
languageRussian	-0.53	0.49	[-1.49, 0.44]	-1.07	0.284	
languageSpanish	0.17	0.47	[-0.76, 1.09]	0.35	0.726	
languageSwahili	0.18	0.48	[-0.76, 1.11]	0.37	0.711	
languageUrdu	0.33	0.49	[-0.63, 1.28]	0.67	0.501	
segment_category.L	1.66	0.08	[1.49, 1.82]	19.72	< .001	***
segment_category.Q	0.31	0.10	[0.12, 0.49]	3.22	0.001	**
segment_category.C	-0.84	0.11	[-1.05, -0.63]	-7.79	< .001	***

Table C1 Fixed-effect estimates from the cumulative link mixed model predicting machine translation quality (0–3).

C2 Model-Level Effects

factor	emmean	SE	CI
Claude Sonnet 4	-2.60	0.14	[-2.87, -2.32]
Cohere Aya Expanse 8B	-0.69	0.14	[-0.96, -0.43]
DeepSeek V3.1	-2.09	0.14	[-2.36, -1.82]
GPT-5	-2.57	0.14	[-2.85, -2.29]
gpt-oss 120b	-1.79	0.14	[-2.06, -1.52]
Llama 4	-1.57	0.14	[-1.84, -1.30]
Mistral Medium 3.1	-2.21	0.14	[-2.49, -1.94]

Table C2 Estimated Marginal Means by Model

contrast	estimate	SE	CI	z_ratio	p_value	Significance
Claude Sonnet 4 - Cohere Aya Expanse 8B	-1.90	0.15	[-2.36, -1.45]	-12.42	< .001	***
Claude Sonnet 4 - DeepSeek V3.1	-0.51	0.15	[-0.96, -0.05]	-3.28	0.018	*
Claude Sonnet 4 - (GPT-5)	-0.02	0.16	[-0.48, 0.43]	-0.15	1.000	
Claude Sonnet 4 - (gpt-oss 120b)	-0.81	0.15	[-1.26, -0.35]	-5.25	< .001	***
Claude Sonnet 4 - Llama 4	-1.03	0.15	[-1.48, -0.57]	-6.68	< .001	***
Claude Sonnet 4 - Mistral Medium 3.1	-0.38	0.15	[-0.84, 0.07]	-2.48	0.165	
Cohere Aya Expanse 8B - DeepSeek V3.1	1.40	0.15	[0.95, 1.84]	9.24	< .001	***
Cohere Aya Expanse 8B - (GPT-5)	1.88	0.15	[1.43, 2.33]	12.31	< .001	***
Cohere Aya Expanse 8B - (gpt-oss 120b)	1.10	0.15	[0.66, 1.54]	7.32	< .001	***
Cohere Aya Expanse 8B - Llama 4	0.88	0.15	[0.43, 1.32]	5.84	< .001	***
Cohere Aya Expanse 8B - Mistral Medium 3.1	1.52	0.15	[1.07, 1.97]	10.03	< .001	***
DeepSeek V3.1 - (GPT-5)	0.48	0.15	[0.03, 0.94]	3.13	0.029	*
DeepSeek V3.1 - (gpt-oss 120b)	-0.30	0.15	[-0.75, 0.15]	-1.97	0.432	
DeepSeek V3.1 - Llama 4	-0.52	0.15	[-0.97, -0.07]	-3.42	0.011	*
DeepSeek V3.1 - Mistral Medium 3.1	0.12	0.15	[-0.33, 0.57]	0.80	0.985	
(GPT-5) - (gpt-oss 120b)	-0.78	0.15	[-1.23, -0.33]	-5.14	< .001	***
(GPT-5) - Llama 4	-1.00	0.15	[-1.45, -0.55]	-6.54	< .001	***
(GPT-5) - Mistral Medium 3.1	-0.36	0.15	[-0.81, 0.09]	-2.34	0.227	
(gpt-oss 120b) - Llama 4	-0.22	0.15	[-0.67, 0.22]	-1.46	0.766	
(gpt-oss 120b) - Mistral Medium 3.1	0.42	0.15	[-0.03, 0.87]	2.78	0.080	
Llama 4 - Mistral Medium 3.1	0.64	0.15	[0.19, 1.09]	4.22	< .001	***

Table C3 Pairwise Model Comparisons (Tukey-adjusted)

C3 Language-Level Effects

factor	emmean	SE	CI
Afrikaans	-1.85	0.34	[-2.52, -1.17]
Arabic	-1.63	0.34	[-2.29, -0.97]
Brazilian Portuguese	-2.76	0.35	[-3.44, -2.09]
Cantonese	-2.06	0.33	[-2.72, -1.41]
Czech	-1.73	0.33	[-2.38, -1.08]
Dutch	-1.55	0.34	[-2.21, -0.89]
Hebrew	-1.68	0.33	[-2.34, -1.03]
Hindi	-1.24	0.33	[-1.89, -0.60]
Japanese	-2.14	0.34	[-2.80, -1.48]
Korean	-1.71	0.34	[-2.38, -1.03]
Mandarin	-3.37	0.36	[-4.07, -2.67]
Russian	-2.37	0.35	[-3.06, -1.68]
Spanish	-1.68	0.33	[-2.33, -1.03]
Swahili	-1.67	0.33	[-2.32, -1.01]
Urdu	-1.52	0.35	[-2.20, -0.84]

Table C4 Estimated Marginal Means by Language

contrast	estimate	SE	CI	z_ratio	p_value	Significance
Afrikaans - Arabic	-0.22	0.48	[-1.84, 1.41]	-0.45	1.000	
Afrikaans - Brazilian Portuguese	0.92	0.48	[-0.73, 2.56]	1.89	0.857	
Afrikaans - Cantonese	0.22	0.48	[-1.40, 1.84]	0.45	1.000	
Afrikaans - Czech	-0.12	0.48	[-1.73, 1.49]	-0.25	1.000	
Afrikaans - Dutch	-0.29	0.48	[-1.91, 1.33]	-0.62	1.000	
Afrikaans - Hebrew	-0.16	0.48	[-1.78, 1.46]	-0.34	1.000	
Afrikaans - Hindi	-0.60	0.47	[-2.21, 1.00]	-1.28	0.995	
Afrikaans - Japanese	0.29	0.48	[-1.33, 1.91]	0.61	1.000	
Afrikaans - Korean	-0.14	0.48	[-1.78, 1.50]	-0.29	1.000	
Afrikaans - Mandarin	1.53	0.49	[-0.15, 3.20]	3.09	0.120	
Afrikaans - Russian	0.53	0.49	[-1.14, 2.19]	1.07	0.999	
Afrikaans - Spanish	-0.17	0.47	[-1.77, 1.44]	-0.35	1.000	
Afrikaans - Swahili	-0.18	0.48	[-1.80, 1.44]	-0.37	1.000	
Afrikaans - Urdu	-0.33	0.49	[-1.98, 1.32]	-0.67	1.000	
Arabic - Brazilian Portuguese	1.13	0.48	[-0.48, 2.75]	2.38	0.532	
Arabic - Cantonese	0.43	0.47	[-1.16, 2.03]	0.92	1.000	
Arabic - Czech	0.10	0.47	[-1.49, 1.68]	0.20	1.000	
Arabic - Dutch	-0.08	0.47	[-1.67, 1.51]	-0.17	1.000	
Arabic - Hebrew	0.05	0.47	[-1.53, 1.64]	0.12	1.000	
Arabic - Hindi	-0.39	0.46	[-1.96, 1.18]	-0.84	1.000	
Arabic - Japanese	0.51	0.47	[-1.09, 2.11]	1.08	0.999	
Arabic - Korean	0.07	0.48	[-1.54, 1.69]	0.16	1.000	
Arabic - Mandarin	1.74	0.49	[0.08, 3.41]	3.56	0.029	*
Arabic - Russian	0.74	0.49	[-0.91, 2.39]	1.52	0.973	
Arabic - Spanish	0.05	0.46	[-1.52, 1.62]	0.11	1.000	
Arabic - Swahili	0.04	0.47	[-1.56, 1.64]	0.08	1.000	
Arabic - Urdu	-0.11	0.48	[-1.74, 1.52]	-0.23	1.000	
Brazilian Portuguese - Cantonese	-0.70	0.48	[-2.32, 0.92]	-1.47	0.980	
Brazilian Portuguese - Czech	-1.04	0.47	[-2.65, 0.57]	-2.19	0.672	
Brazilian Portuguese - Dutch	-1.21	0.48	[-2.83, 0.40]	-2.54	0.409	
Brazilian Portuguese - Hebrew	-1.08	0.48	[-2.70, 0.54]	-2.27	0.616	
Brazilian Portuguese - Hindi	-1.52	0.47	[-3.12, 0.08]	-3.23	0.082	
Brazilian Portuguese - Japanese	-0.63	0.48	[-2.24, 0.99]	-1.31	0.993	
Brazilian Portuguese - Korean	-1.06	0.48	[-2.70, 0.58]	-2.19	0.672	
Brazilian Portuguese - Mandarin	0.61	0.49	[-1.07, 2.29]	1.23	0.996	
Brazilian Portuguese - Russian	-0.39	0.49	[-2.06, 1.27]	-0.80	1.000	
Brazilian Portuguese - Spanish	-1.08	0.47	[-2.69, 0.52]	-2.29	0.596	
Brazilian Portuguese - Swahili	-1.10	0.48	[-2.72, 0.53]	-2.29	0.596	
Brazilian Portuguese - Urdu	-1.25	0.49	[-2.90, 0.41]	-2.56	0.398	
Cantonese - Czech	-0.34	0.47	[-1.92, 1.25]	-0.72	1.000	
Cantonese - Dutch	-0.51	0.47	[-2.10, 1.08]	-1.09	0.999	
Cantonese - Hebrew	-0.38	0.47	[-1.97, 1.21]	-0.81	1.000	
Cantonese - Hindi	-0.82	0.46	[-2.40, 0.76]	-1.76	0.912	
Cantonese - Japanese	0.08	0.47	[-1.52, 1.67]	0.16	1.000	
Cantonese - Korean	-0.36	0.48	[-1.97, 1.26]	-0.75	1.000	
Cantonese - Mandarin	1.31	0.49	[-0.34, 2.96]	2.69	0.310	
Cantonese - Russian	0.31	0.48	[-1.33, 1.95]	0.64	1.000	
Cantonese - Spanish	-0.38	0.47	[-1.96, 1.20]	-0.82	1.000	
Cantonese - Swahili	-0.39	0.47	[-1.99, 1.20]	-0.84	1.000	
Cantonese - Urdu	-0.54	0.48	[-2.17, 1.08]	-1.13	0.999	
Czech - Dutch	-0.17	0.47	[-1.75, 1.41]	-0.37	1.000	
Czech - Hebrew	-0.04	0.47	[-1.62, 1.54]	-0.09	1.000	
Czech - Hindi	-0.48	0.46	[-2.05, 1.08]	-1.05	0.999	

Table C5 Pairwise Language Comparisons (Tukey-adjusted) *continued on next page*

Czech - Japanese	0.41	0.47	[-1.17, 2.00]	0.88	1.000	
Czech - Korean	-0.02	0.47	[-1.63, 1.59]	-0.04	1.000	
Czech - Mandarin	1.65	0.49	[0.00, 3.30]	3.39	0.050	*
Czech - Russian	0.65	0.48	[-0.99, 2.28]	1.34	0.992	
Czech - Spanish	-0.05	0.46	[-1.61, 1.52]	-0.10	1.000	
Czech - Swahili	-0.06	0.47	[-1.64, 1.53]	-0.12	1.000	
Czech - Urdu	-0.21	0.48	[-1.82, 1.41]	-0.43	1.000	
Dutch - Hebrew	0.13	0.47	[-1.46, 1.72]	0.28	1.000	
Dutch - Hindi	-0.31	0.46	[-1.88, 1.26]	-0.67	1.000	
Dutch - Japanese	0.59	0.47	[-1.01, 2.18]	1.25	0.996	
Dutch - Korean	0.15	0.48	[-1.46, 1.77]	0.32	1.000	
Dutch - Mandarin	1.82	0.49	[0.17, 3.48]	3.73	0.016	*
Dutch - Russian	0.82	0.48	[-0.82, 2.46]	1.69	0.936	
Dutch - Spanish	0.13	0.46	[-1.45, 1.70]	0.28	1.000	
Dutch - Swahili	0.12	0.47	[-1.48, 1.71]	0.25	1.000	
Dutch - Urdu	-0.03	0.48	[-1.66, 1.59]	-0.07	1.000	
Hebrew - Hindi	-0.44	0.46	[-2.01, 1.13]	-0.95	1.000	
Hebrew - Japanese	0.45	0.47	[-1.14, 2.05]	0.97	1.000	
Hebrew - Korean	0.02	0.48	[-1.59, 1.63]	0.04	1.000	
Hebrew - Mandarin	1.69	0.49	[0.03, 3.34]	3.46	0.040	*
Hebrew - Russian	0.69	0.48	[-0.95, 2.33]	1.42	0.986	
Hebrew - Spanish	0.00	0.46	[-1.58, 1.57]	-0.01	1.000	
Hebrew - Swahili	-0.02	0.47	[-1.61, 1.58]	-0.03	1.000	
Hebrew - Urdu	-0.17	0.48	[-1.79, 1.46]	-0.35	1.000	
Hindi - Japanese	0.90	0.47	[-0.68, 2.47]	1.93	0.840	
Hindi - Korean	0.46	0.47	[-1.13, 2.06]	0.98	1.000	
Hindi - Mandarin	2.13	0.48	[0.49, 3.77]	4.40	0.001	**
Hindi - Russian	1.13	0.48	[-0.50, 2.76]	2.35	0.551	
Hindi - Spanish	0.44	0.46	[-1.12, 1.99]	0.95	1.000	
Hindi - Swahili	0.43	0.47	[-1.15, 2.01]	0.92	1.000	
Hindi - Urdu	0.28	0.47	[-1.33, 1.89]	0.58	1.000	
Japanese - Korean	-0.43	0.48	[-2.05, 1.18]	-0.91	1.000	
Japanese - Mandarin	1.23	0.49	[-0.42, 2.89]	2.53	0.416	
Japanese - Russian	0.23	0.48	[-1.41, 1.87]	0.48	1.000	
Japanese - Spanish	-0.46	0.47	[-2.04, 1.12]	-0.98	1.000	
Japanese - Swahili	-0.47	0.47	[-2.07, 1.13]	-1.00	1.000	
Japanese - Urdu	-0.62	0.48	[-2.25, 1.01]	-1.29	0.994	
Korean - Mandarin	1.67	0.49	[-0.01, 3.34]	3.38	0.052	
Korean - Russian	0.67	0.49	[-1.00, 2.33]	1.36	0.991	
Korean - Spanish	-0.02	0.47	[-1.63, 1.58]	-0.05	1.000	
Korean - Swahili	-0.04	0.48	[-1.65, 1.58]	-0.08	1.000	
Korean - Urdu	-0.19	0.49	[-1.83, 1.46]	-0.38	1.000	
Mandarin - Russian	-1.00	0.50	[-2.69, 0.69]	-2.01	0.793	
Mandarin - Spanish	-1.69	0.48	[-3.34, -0.05]	-3.50	0.036	*
Mandarin - Swahili	-1.70	0.49	[-3.36, -0.05]	-3.50	0.035	*
Mandarin - Urdu	-1.86	0.50	[-3.54, -0.17]	-3.74	0.015	*
Russian - Spanish	-0.69	0.48	[-2.32, 0.94]	-1.44	0.984	
Russian - Swahili	-0.70	0.48	[-2.34, 0.94]	-1.45	0.982	
Russian - Urdu	-0.85	0.49	[-2.52, 0.82]	-1.73	0.923	
Spanish - Swahili	-0.01	0.47	[-1.59, 1.57]	-0.02	1.000	
Spanish - Urdu	-0.16	0.48	[-1.77, 1.45]	-0.34	1.000	
Swahili - Urdu	-0.15	0.48	[-1.78, 1.48]	-0.31	1.000	

Table C5 Pairwise Language Comparisons (Tukey-adjusted)

C4 Segment Category Effects

factor	emmean	SE	CI
cultural concepts	-2.70	0.10	[-2.90, -2.50]
holidays	-3.02	0.14	[-3.28, -2.75]
idioms	-1.15	0.14	[-1.43, -0.88]
puns	-0.85	0.13	[-1.10, -0.60]

Table C6 Estimated Marginal Means by Segment Category

contrast	estimate	SE	CI	z_ratio	p_value	Significance
cultural concepts - holidays	0.31	0.12	[0.01, 0.62]	2.67	0.039	*
cultural concepts - idioms	-1.55	0.13	[-1.88, -1.22]	-12.13	< .001	***
cultural concepts - puns	-1.85	0.11	[-2.14, -1.56]	-16.42	< .001	***
holidays - idioms	-1.86	0.16	[-2.27, -1.46]	-11.90	< .001	***
holidays - puns	-2.16	0.14	[-2.54, -1.79]	-14.95	< .001	***
idioms - puns	-0.30	0.15	[-0.69, 0.09]	-2.00	0.188	

Table C7 Pairwise Category Comparisons (Tukey-adjusted)

C5 Inter-Rater Reliability

Inter-rater reliability (IRR) was assessed to evaluate the consistency of human ratings of translation quality across participants. Because ratings were ordinal (e.g., ranging from “very good / nearly perfect” to “serious failures”) and involved multiple raters, we selected complementary reliability measures to capture different aspects of agreement.

We report Krippendorff’s α (ordinal), which is designed for ordered categorical data and is robust to missing values, providing a single coefficient reflecting agreement beyond chance. We additionally report Gwet’s AC2 with quadratic weights, which accounts for chance agreement while being less sensitive to prevalence and marginal distributions than Cohen’s κ . Quadratic weights penalise larger disagreements more heavily, reflecting the

ordered structure of the rating scale. Observed and expected agreement rates derived from AC2 are also reported to aid interpretation of reliability in terms of raw concordance.

Ratings corresponding to “segment not translated” (NA) were excluded from all IRR calculations, as they reflect missing or invalid quality judgments rather than graded assessments. IRR was computed at multiple levels, including overall agreement across all languages, models, and segment categories, as well as stratified by language, model, and segment category (cultural concepts, holidays, idioms, and puns).

IRR calculations were based on item $\times$ rater matrices constructed from the cleaned data and were implemented in R using the `irr` and `irrCAC` packages.

metric	estimate	lower 95 ci	upper 95 ci	observed agreement	expected agreement	scope	language	model	segment category
Krippendorff_alpha	0.448197144	NA	NA	NA	NA	Overall	NA	NA	NA
Gwet_AC1_weighted	0.41225	(0.31,0.514)	NA	0.755857523	0.584613092	Overall	NA	NA	NA
Krippendorff_alpha	0.498850973	NA	NA	NA	NA	Afrikaans	Afrikaans	NA	NA
Gwet_AC1_weighted	0.14534	(-0.098,0.389)	NA	0.632263084	0.569726302	Afrikaans	Afrikaans	NA	NA
Krippendorff_alpha	0.551735695	NA	NA	NA	NA	Arabic	Arabic	NA	NA
Gwet_AC1_weighted	0.61952	(0.193,1)	NA	0.849890557	0.605471591	Arabic	Arabic	NA	NA
Krippendorff_alpha	0.354424333	NA	NA	NA	NA	Brazilian Portuguese	Brazilian Portuguese	NA	NA
Gwet_AC1_weighted	0.56154	(0.169,0.955)	NA	0.83130482	0.615255895	Brazilian Portuguese	Brazilian Portuguese	NA	NA
Krippendorff_alpha	0.386155192	NA	NA	NA	NA	Cantonese	Cantonese	NA	NA
Gwet_AC1_weighted	0.58182	(0.14,1)	NA	0.824587744	0.580530558	Cantonese	Cantonese	NA	NA
Krippendorff_alpha	0.501678657	NA	NA	NA	NA	Czech	Czech	NA	NA
Gwet_AC1_weighted	0.41474	(-0.014,0.843)	NA	0.731120638	0.540580132	Czech	Czech	NA	NA
Krippendorff_alpha	0.57461557	NA	NA	NA	NA	Dutch	Dutch	NA	NA
Gwet_AC1_weighted	0.55692	(0.085,1)	NA	0.768596935	0.477734454	Dutch	Dutch	NA	NA
Krippendorff_alpha	0.525872162	NA	NA	NA	NA	Hebrew	Hebrew	NA	NA
Gwet_AC1_weighted	0.52	(0.037,1)	NA	0.788096253	0.558531884	Hebrew	Hebrew	NA	NA
Krippendorff_alpha	0.269765185	NA	NA	NA	NA	Hindi	Hindi	NA	NA
Gwet_AC1_weighted	0.64013	(0.201,1)	NA	0.833581517	0.537553424	Hindi	Hindi	NA	NA
Krippendorff_alpha	0.476073172	NA	NA	NA	NA	Japanese	Japanese	NA	NA
Gwet_AC1_weighted	0.42556	(0,0.851)	NA	0.780653592	0.618154078	Japanese	Japanese	NA	NA
Krippendorff_alpha	0.512877424	NA	NA	NA	NA	Korean	Korean	NA	NA
Gwet_AC1_weighted	0.29641	(-0.134,0.727)	NA	0.770063675	0.673197163	Korean	Korean	NA	NA
Krippendorff_alpha	0.267738681	NA	NA	NA	NA	Mandarin	Mandarin	NA	NA
Gwet_AC1_weighted	0.60908	(0.341,0.877)	NA	0.806393163	0.504740189	Mandarin	Mandarin	NA	NA
Krippendorff_alpha	0.372753672	NA	NA	NA	NA	Russian	Russian	NA	NA
Gwet_AC1_weighted	0.3071	(-0.202,0.817)	NA	0.745234394	0.63231795	Russian	Russian	NA	NA
Krippendorff_alpha	0.375234491	NA	NA	NA	NA	Spanish	Spanish	NA	NA
Gwet_AC1_weighted	0.52538	(0.028,1)	NA	0.808558288	0.596638428	Spanish	Spanish	NA	NA
Krippendorff_alpha	0.485056033	NA	NA	NA	NA	Swahili	Swahili	NA	NA
Gwet_AC1_weighted	0.24906	(-0.315,0.813)	NA	0.714105052	0.619284773	Swahili	Swahili	NA	NA
Krippendorff_alpha	0.386782145	NA	NA	NA	NA	Urdu	Urdu	NA	NA
Gwet_AC1_weighted	0.13664	(-0.029,0.302)	NA	0.641737452	0.585034893	Urdu	Urdu	NA	NA

Table C8 Inter-rater reliability statistics for segment-level MT quality ratings *continued on next page*

Krippendorff_ alpha	0.362971562	NA	NA	NA	NA	Claude Sonnet 4	NA	Claude Sonnet 4	NA
Gwet_AC1_ weighted	0.42987	(0.333,0.527)	NA	0.778941763	0.612268497	Claude Sonnet 4	NA	Claude Sonnet 4	NA
Krippendorff_ alpha	0.591731477	NA	NA	NA	NA	Cohere Aya Expanse 8B	NA	Cohere Aya Expanse 8B	NA
Gwet_AC1_ weighted	0.22705	(0.112,0.342)	NA	0.730289925	0.651062192	Cohere Aya Expanse 8B	NA	Cohere Aya Expanse 8B	NA
Krippendorff_ alpha	0.365021429	NA	NA	NA	NA	DeepSeek V3.1	NA	DeepSeek V3.1	NA
Gwet_AC1_ weighted	0.23255	(0.142,0.323)	NA	0.709368798	0.621304321	DeepSeek V3.1	NA	DeepSeek V3.1	NA
Krippendorff_ alpha	0.390678454	NA	NA	NA	NA	GPT-5	NA	GPT-5	NA
Gwet_AC1_ weighted	0.42612	(0.325,0.527)	NA	0.778789507	0.614534099	GPT-5	NA	GPT-5	NA
Krippendorff_ alpha	0.425609176	NA	NA	NA	NA	gpt-oss 120B	NA	gpt-oss 120B	NA
Gwet_AC1_ weighted	0.11716	(0.057,0.178)	NA	0.688983685	0.647708789	gpt-oss 120B	NA	gpt-oss 120B	NA
Krippendorff_ alpha	0.492022726	NA	NA	NA	NA	Llama 4	NA	Llama 4	NA
Gwet_AC1_ weighted	0.1872	(0.103,0.271)	NA	0.715267408	0.649690855	Llama 4	NA	Llama 4	NA
Krippendorff_ alpha	0.353854999	NA	NA	NA	NA	Mistral Medium 3.1	NA	Mistral Medium 3.1	NA
Gwet_AC1_ weighted	0.29617	(0.209,0.384)	NA	0.747983902	0.641936198	Mistral Medium 3.1	NA	Mistral Medium 3.1	NA
Krippendorff_ alpha	0.441472615	NA	NA	NA	NA	cultural concepts	NA	NA	cultural concepts
Gwet_AC1_ weighted	0.34828	(0.28,0.417)	NA	0.745008769	0.608740413	cultural concepts	NA	NA	cultural concepts
Krippendorff_ alpha	0.380075728	NA	NA	NA	NA	holidays	NA	NA	holidays
Gwet_AC1_ weighted	0.40557	(0.305,0.507)	NA	0.733721118	0.552041014	holidays	NA	NA	holidays
Krippendorff_ alpha	0.404880664	NA	NA	NA	NA	idioms	NA	NA	idioms
Gwet_AC1_ weighted	0.10721	(0.048,0.166)	NA	0.737554455	0.706039385	idioms	NA	NA	idioms
Krippendorff_ alpha	0.307338984	NA	NA	NA	NA	puns	NA	NA	puns
Gwet_AC1_ weighted	0.26271	(0.16,0.366)	NA	0.757788673	0.671483717	puns	NA	NA	puns

Table C8 Inter-rater reliability statistics for segment-level MT quality ratings

language	model	alpha	ac2	pairwise_ agree	strict_ agree	n_ items	n_ raters
Arabic	Cohere Aya Expans 8B	0.674813037	NA	55.1	26.1	23	5
Japanese	Llama 4	0.65637168	NA	56.7	36.8	20	5
Czech	Llama 4	0.65207732	NA	53.9	24	25	5
Hebrew	gpt-oss 120B	0.644062377	NA	52.4	20	25	5
Arabic	Llama 4	0.634999976	NA	56.1	24	25	5
Arabic	Claude Sonnet 4	0.634900605	NA	64.7	36	25	5
Urdu	Cohere Aya Expans 8B	0.617955706	NA	52.9	50	19	5
Hebrew	DeepSeek V3.1	0.614170634	NA	63.8	40	25	5
Hebrew	Cohere Aya Expans 8B	0.611726618	NA	57.3	17.4	23	5
Japanese	Cohere Aya Expans 8B	0.584984776	NA	54.5	21.7	23	5
Dutch	Cohere Aya Expans 8B	0.577023671	NA	51.4	31.6	24	5
Dutch	gpt-oss 120B	0.564995102	NA	57.5	27.3	25	5
Dutch	Claude Sonnet 4	0.555536604	NA	59.5	36.4	25	5
Cantonese	Llama 4	0.550254155	NA	45.3	17.4	23	5
Arabic	Mistral Medium 3.1	0.544804854	NA	60	32	25	5
Czech	gpt-oss 120B	0.528554281	NA	53.2	34.8	25	5
Hebrew	Llama 4	0.518538232	NA	49	25	24	5
Arabic	gpt-oss 120B	0.513416055	NA	50.4	12	25	5
Korean	GPT-5	0.512390998	NA	45.7	8	25	5
Czech	Cohere Aya Expans 8B	0.510046027	NA	47.4	14.3	22	5
Hebrew	Claude Sonnet 4	0.502641466	NA	47.5	28	25	5
Dutch	DeepSeek V3.1	0.493616221	NA	52.7	23.8	25	5
Dutch	Mistral Medium 3.1	0.493573969	NA	61.8	40.9	25	5
Korean	DeepSeek V3.1	0.486751851	NA	45	16	25	5
Arabic	GPT-5	0.481986498	NA	59.2	32	25	5
Korean	Cohere Aya Expans 8B	0.474465656	NA	45.6	20	25	5
Spanish	Llama 4	0.47124898	NA	45.1	12.5	25	5
Dutch	Llama 4	0.47021559	NA	55.2	28.6	25	5
Czech	Claude Sonnet 4	0.457016233	NA	59.7	37.5	24	5
Afrikaans	GPT-5	0.456513385	NA	71.4	47.6	24	5
Russian	Cohere Aya Expans 8B	0.448616905	NA	44.4	20	25	5
Afrikaans	gpt-oss 120B	0.441137352	NA	61.1	36.4	24	5
Cantonese	Cohere Aya Expans 8B	0.439844702	NA	38.1	16	25	5
Korean	Llama 4	0.432970093	NA	47.8	8.7	25	5
Czech	GPT-5	0.428335745	NA	59.6	36	25	5
Brazilian Portuguese	gpt-oss 120B	0.426784937	NA	62	34.8	25	5
Korean	gpt-oss 120B	0.411637737	NA	43.6	8	25	5
Russian	gpt-oss 120B	0.402388898	NA	41.6	20	25	5
Afrikaans	Cohere Aya Expans 8B	0.402248913	NA	41.7	27.3	23	5
Brazilian Portuguese	DeepSeek V3.1	0.400921077	NA	58.1	36	25	5
Mandarin	Llama 4	0.396096645	NA	54.1	28	25	5
Swahili	Llama 4	0.392872584	NA	52.9	30.4	25	5
Czech	Mistral Medium 3.1	0.391101109	NA	55.8	20.8	25	5
Brazilian Portuguese	Claude Sonnet 4	0.389526749	NA	64	44	25	5

Table C9 Inter-rater reliability statistics for holistic text MT quality ratings *continued on next page*

Spanish	GPT-5	0.384567319	NA	48.2	20	25	5
Hebrew	GPT-5	0.382856402	NA	57.6	28	25	5
Japanese	gpt-oss 120B	0.382726192	NA	51.8	28	25	5
Russian	DeepSeek V3.1	0.38161071	NA	46.8	17.4	24	5
Hindi	Mistral Medium 3.1	0.380938245	NA	38.5	8	25	5
Korean	Claude Sonnet 4	0.377620246	NA	46.6	12	25	5
Cantonese	Mistral Medium 3.1	0.371804013	NA	44.8	16	25	5
Cantonese	GPT-5	0.360470042	NA	48.6	20	25	5
Cantonese	gpt-oss 120B	0.348227295	NA	45.5	20.8	24	5
Brazilian Portuguese	Cohere Aya Expanse 8B	0.347385294	NA	54.6	29.2	24	5
Mandarin	Cohere Aya Expanse 8B	0.345099047	NA	46.1	16.7	24	5
Russian	Llama 4	0.333786874	NA	41.6	12	25	5
Brazilian Portuguese	Llama 4	0.321316883	NA	51	20	25	5
Hindi	Cohere Aya Expanse 8B	0.318208174	NA	46.8	18.2	23	5
Afrikaans	Claude Sonnet 4	0.306930278	NA	63	34.8	25	5
Arabic	DeepSeek V3.1	0.306619915	NA	44.7	12	25	5
Hindi	GPT-5	0.305698654	NA	48.3	20	25	5
Spanish	Claude Sonnet 4	0.301239732	NA	47.4	12	25	5
Czech	DeepSeek V3.1	0.29960442	NA	52.3	24	25	5
Urdu	GPT-5	0.288346858	NA	51.7	29.4	18	5
Urdu	Claude Sonnet 4	0.282773726	NA	67.1	50	20	5
Spanish	gpt-oss 120B	0.281928166	NA	43	13	25	5
Mandarin	gpt-oss 120B	0.278237639	NA	57.2	24	25	5
Russian	Mistral Medium 3.1	0.277913363	NA	49.2	20.8	25	5
Japanese	Mistral Medium 3.1	0.270812278	NA	60.1	33.3	24	5
Swahili	Mistral Medium 3.1	0.270118901	NA	43.1	16	25	5
Japanese	Claude Sonnet 4	0.263590307	NA	58.6	28	25	5
Swahili	Claude Sonnet 4	0.2617325	NA	52.4	28	25	5
Spanish	DeepSeek V3.1	0.257918185	NA	42.4	8	25	5
Russian	Claude Sonnet 4	0.257401126	NA	55.1	28	25	5
Spanish	Cohere Aya Expanse 8B	0.257126966	NA	41.9	13	25	5
Afrikaans	Mistral Medium 3.1	0.256137166	NA	47.8	25	24	5
Cantonese	Claude Sonnet 4	0.251197556	NA	51.4	24	25	5
Hindi	Llama 4	0.24301364	NA	43.5	16	25	5
Korean	Mistral Medium 3.1	0.242460239	NA	48.1	12.5	24	5
Hebrew	Mistral Medium 3.1	0.239619599	NA	44.8	20	25	5
Afrikaans	DeepSeek V3.1	0.232375967	NA	55.7	37.5	25	5
Hindi	DeepSeek V3.1	0.229249261	NA	41	12	25	5
Brazilian Portuguese	Mistral Medium 3.1	0.212810949	NA	59	32	25	5
Urdu	Llama 4	0.209767274	NA	54.4	35	20	5
Cantonese	DeepSeek V3.1	0.208166722	NA	50.9	16	25	5
Swahili	gpt-oss 120B	0.201556295	NA	38.8	21.7	25	5
Dutch	GPT-5	0.201535135	NA	52.2	26.1	25	5
Hindi	gpt-oss 120B	0.195261741	NA	41.8	12	25	5
Spanish	Mistral Medium 3.1	0.194733517	NA	47.6	20	25	5

Table C9 Inter-rater reliability statistics for holistic text MT quality ratings *continued on next page*

Brazilian Portuguese	GPT-5	0.189900439	NA	64.3	32	25	5
Japanese	GPT-5	0.181908943	NA	59.3	29.2	24	5
Urdu	Mistral Medium 3.1	0.179128246	NA	54.4	25	20	5
Japanese	DeepSeek V3.1	0.17029338	NA	47.9	20	25	5
Swahili	GPT-5	0.168215451	NA	60.1	37.5	24	5
Hindi	Claude Sonnet 4	0.165242137	NA	38.1	8	25	5
Swahili	DeepSeek V3.1	0.139262956	NA	49.6	16	25	5
Swahili	Cohere Aya Expanse 8B	0.136498089	NA	73	33.3	25	5
Urdu	gpt-oss 120B	0.128582875	NA	38	10.5	20	5
Afrikaans	Llama 4	0.121541552	NA	55.2	21.7	23	5
Mandarin	DeepSeek V3.1	0.114767658	NA	65.1	35	20	5
Russian	GPT-5	0.087226249	NA	48	24	25	5
Mandarin	Mistral Medium 3.1	0.080766145	NA	60.3	32	25	5
Mandarin	Claude Sonnet 4	0.072938315	NA	52.4	20	25	5
Urdu	DeepSeek V3.1	0.017972445	NA	52.1	30	20	5

Table C9 Inter-rater reliability statistics for holistic text MT quality ratings